# STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 9 -

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# ANALYSIS
# of
# STATICALLY DETERMINATE SYSTEMS UNDER LIVE LOADS

In the previous lessons we developed techniques for analyzing the forces in structural members due to dead or fixed loads.

If a structure is subjected to a live or moving load, the variation of the shear and bending moment in the member best described using the influence line.

An influence line represents the variation of either the reaction, shear, moment, or deflection at a specific point in a member as a concentrated force moves over the member.

By means of the inluence line, we can tell where the moving load should be placed so that it creates the greatest influence at the specified point.

The magnitude of the associated reaction, shear, moment, or deflection at the point can then be calculated from the ordinates of the influence-line diagram.

For these reasons, influence lines plan an important part in the design of bridges, industrial crane rails, conveyors, and other structures where loads move across their span.

Although the procedure for constructing an influence line is rather basic, one should clearly be aware of the difference between constructing an influence line and constructing a shear or moment diagram.

- **Influence lines represent the effect of a moving load only at a specified point on a member**
- **Shear and moment diagrams represent the effect of fixed loads at all points along the axis of the member**

# Constructing Influence Lines

Unit load applied to the structure is assumed as a fixed load at a distance x with respect to a reference point and the functional form of the influence line for the parameter of interest is derived.

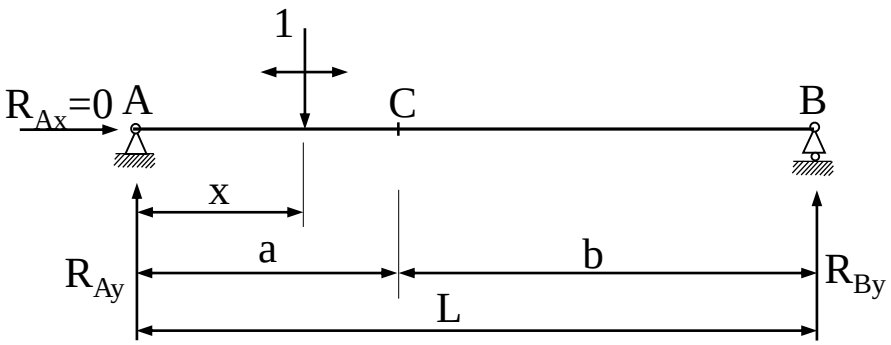
In case of statically determinate systems, the influence lines of internal loads and support reactions are linear. Therefore it is sufficient to calculate the ordinates of influence lines at a few critical sections.

# *Unit vertical load*

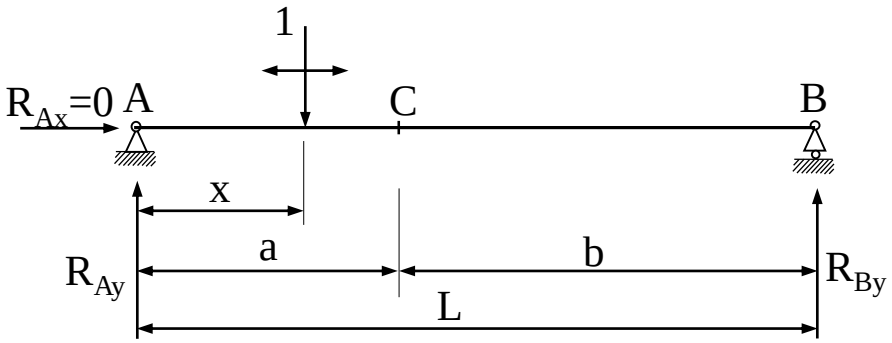
Let's draw the influence lines for

- support reactions
- shear force at point C
- bending moment at point C

of a simply supported beam.



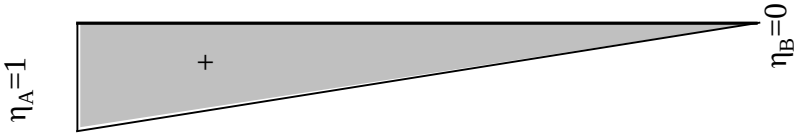
# Influence line of vertical reaction of support A




$$\sum M_B = 0 \qquad R_{Ay} = 1.(L - x)/L$$

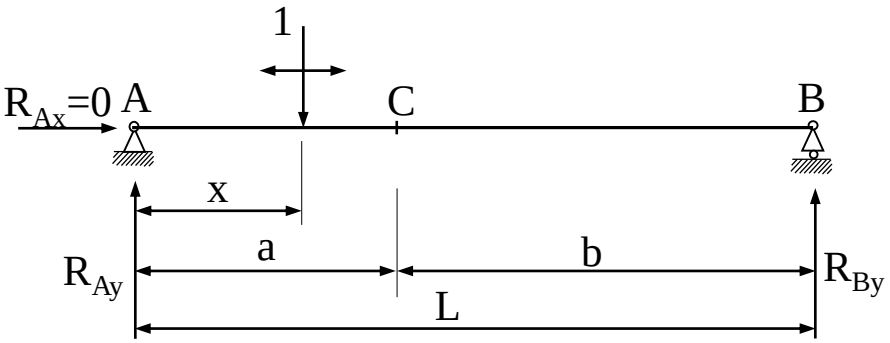
Influence line function of vertical reaction of support A: $\boldsymbol{\eta R_{Ay} = (L - x)/L}$

$$\boldsymbol{x = 0 \rightarrow \eta R_{Ay,A} = (L - 0)/L = 1}$$
$$\boldsymbol{x = L \rightarrow \eta R_{Ay,B} = (L - L)/L = 0}$$




$\boldsymbol{\eta R_{Ay}}$

# Influence line of vertical reaction of support B

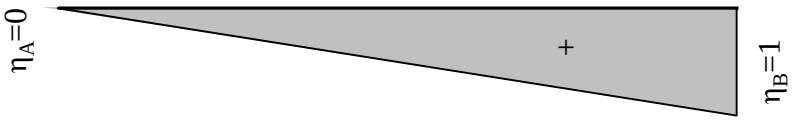



$$\sum M_A = 0 \qquad R_{By} = 1.x/L$$

Influence line function of vertical reaction of support A: $\boldsymbol{\eta R_{By} = x/L}$

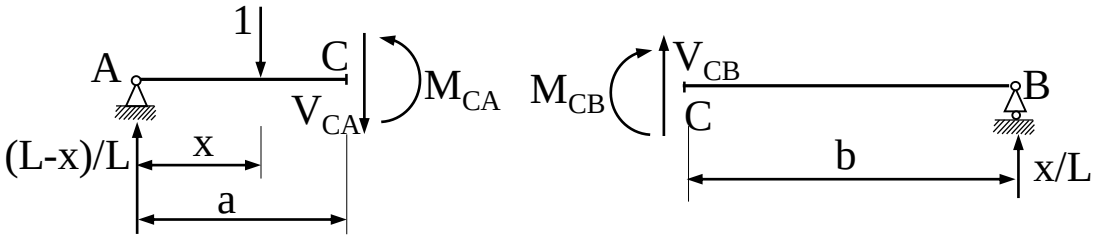
$$\boldsymbol{x = 0 \rightarrow \eta R_{By,A} = (0)/L = 0}$$

$$\boldsymbol{x = L \rightarrow \eta R_{By,B} = L/L = 1}$$




$\boldsymbol{\eta R_{By}}$

# Influence line of shear force at C




*The load is between A and C (0<x<a)*

Consider CB part since there is less load acting on CB

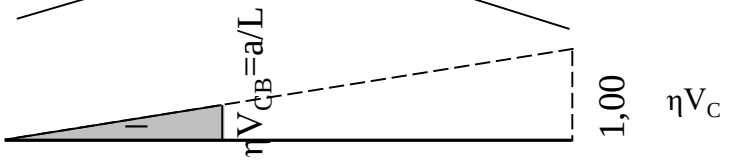
$$\sum P_y = 0 \qquad V_{CB} = -R_{By} = -1.x/L$$

Influence line function of shear force at point C: $\eta V_{CB} = -\eta R_{By} = -x/L$

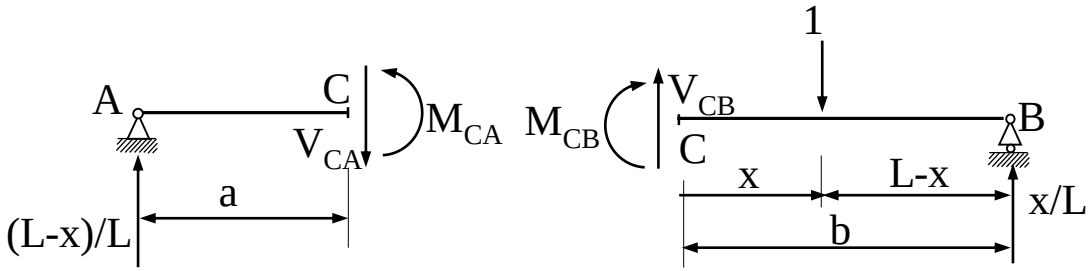
$$x = 0 \rightarrow \eta V_{CB,A} = 0$$

$$x = a \rightarrow \eta V_{CB,C} = -a/L$$

$x = L \rightarrow \eta V_{CB,B} = -1$ → Invalid



# Influence line of shear force at C




*The load is between C and B (a<x<L)*

Consider AC part since there is less load acting on AC

$$\sum P_y = 0 \qquad V_{CA} = R_{Ay} = 1.(\mathrm{L} - x)/L$$

Influence line function of shear force at point C: $\eta V_{CA} = \eta R_{Ay} = (\mathbf{L} - \mathbf{x})/\mathbf{L}$

$$x = a \rightarrow \eta V_{CA,C} = \mathbf{b/L}$$

$$x = L \rightarrow \eta V_{CA,B} = \mathbf{0}$$

$$x = 0 \rightarrow \eta V_{CA,A} = \mathbf{1}$$ → Invalid

# Influence line of shear force at C

*The load is between A and C (0<x<a)*

$$\eta V_{CB} = -\eta R_{By} = -x/L$$

$$x = 0 \rightarrow \eta V_{CB,A} = 0$$

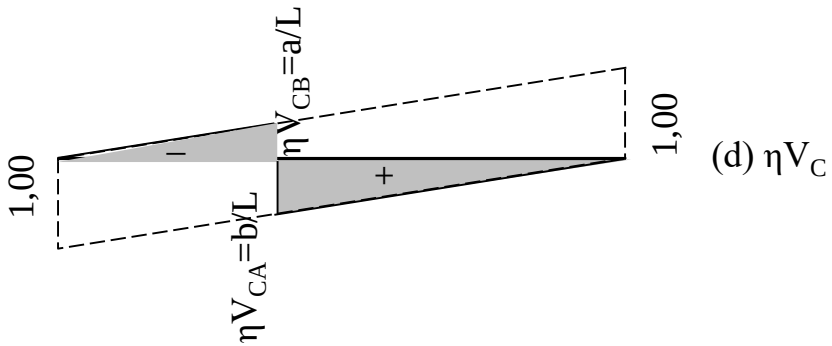
$$x = a \rightarrow \eta V_{CB,C} = -a/L$$

*The load is between C and B (a<x<L)*

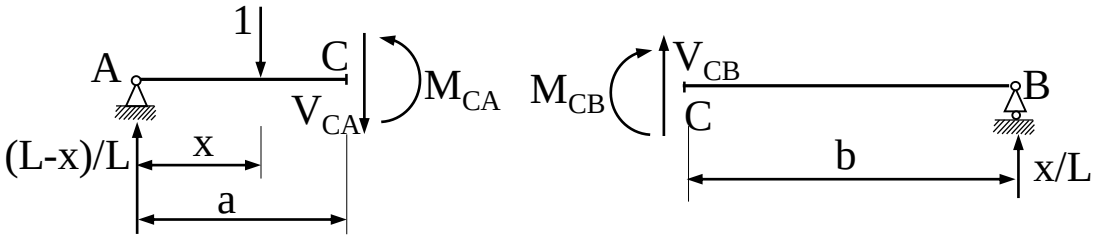
$$\eta V_{CA} = \eta R_{Ay} = (L - x)/L$$

$$x = a \rightarrow \eta V_{CA,C} = b/L$$

$$x = L \rightarrow \eta V_{CA,B} = 0$$



# Influence line of bending moment at C




*The load is between A and C (0<x<a)*

Consider CB part since there is less load acting on CB

$$\sum M_C = 0 \qquad M_{CB} = b.R_{By} = \mathrm{b}.1.x/L$$

Influence line function of bending moment at point C:

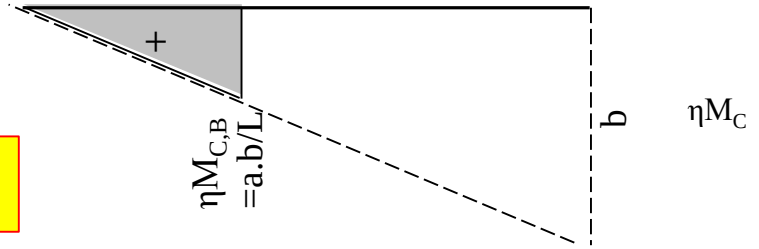
$$\eta M_{CB} = b.\eta R_{By} = \mathbf{b}\left(\frac{\mathbf{x}}{\mathbf{L}}\right)$$

$$x = 0 \rightarrow \eta M_{CB,A} = 0$$

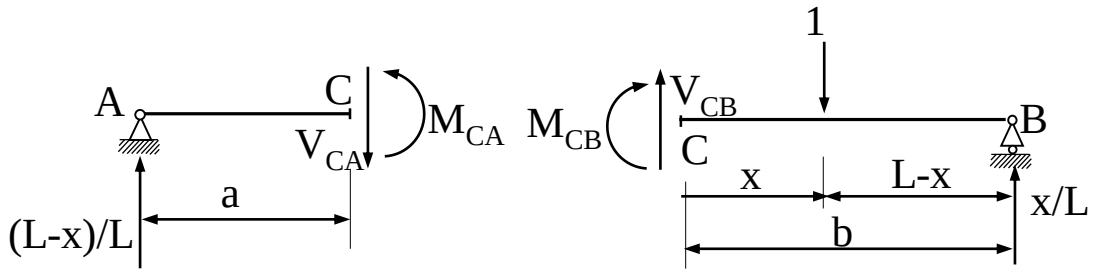
$$x = a \rightarrow \eta M_{CB,C} = \mathbf{a}.\mathbf{b}/\mathbf{L}$$

$$x = L \rightarrow \eta M_{CB,B} = \mathbf{b}$$

Invalid



# Influence line of bending moment at C




*The load is between C and B (a<x<L)*

Consider AC part since there is less load acting on AC

$$\sum M_C = 0 \qquad M_{CA} = a.R_{Ay} = \mathrm{a}.1.(\mathrm{L}-x)/L$$

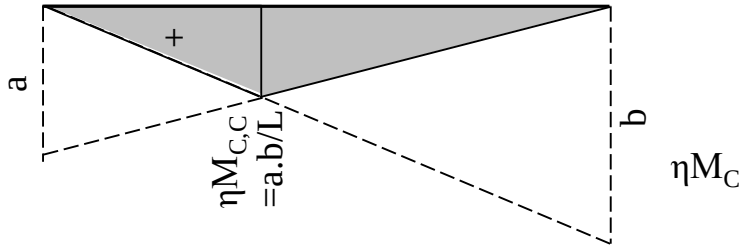
Influence line function of bending moment at point C:

$$\eta \boldsymbol{M_{CA}} = \boldsymbol{a}.\boldsymbol{\eta R_{By}} = \mathbf{a}\left(\frac{\mathbf{L}-\mathbf{x}}{\mathbf{L}}\right)$$

$$\boldsymbol{x = a \rightarrow \eta M_{CA,C} = a.b/L}$$

$$\boldsymbol{x = L \rightarrow \eta M_{CA,B} = 0}$$

$$\boldsymbol{x = 0 \rightarrow \eta M_{CA,A} = a}$$ → Invalid



# Influence line of bending moment at C

*The load is between A and C (0<x<a)*

$$\eta M_{CB} = b.\eta R_{By} = b\left(\frac{x}{L}\right)$$

$$x = 0 \rightarrow \eta M_{CB,A} = 0$$

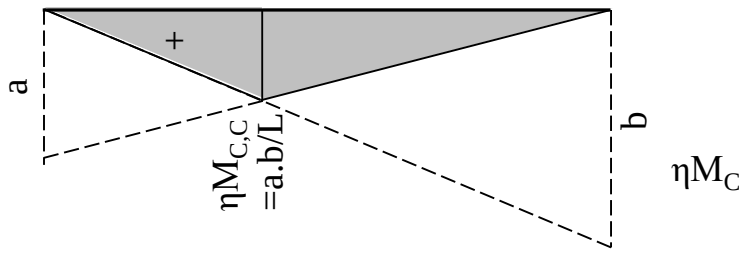
$$x = a \rightarrow \eta M_{CB,C} = a.b/L$$

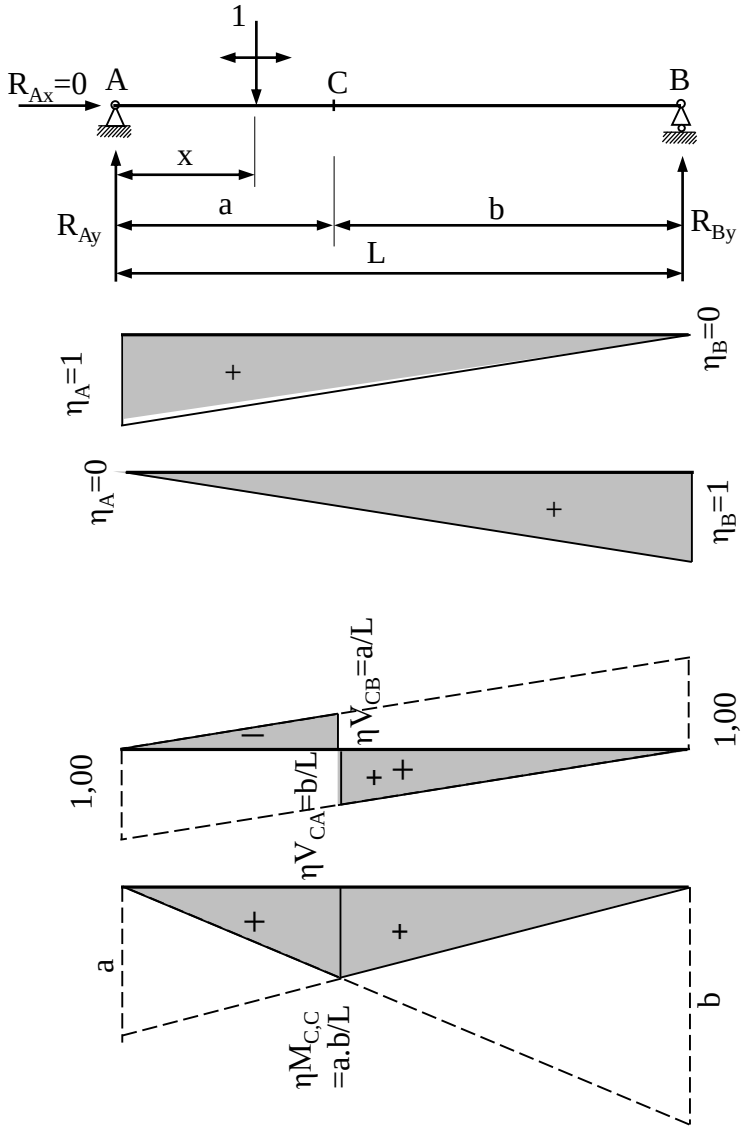
*The load is between C and B (a<x<L)*

$$\eta M_{CA} = a.\eta R_{By} = a\left(\frac{L - x}{L}\right)$$

$$x = a \rightarrow \eta M_{CA,C} = a.b/L$$

$$x = L \rightarrow \eta M_{CA,B} = 0$$






$\eta R_{Ay}$

$$\eta R_{Ay} = (L - x)/L$$

$\eta R_{By}$

$$\eta R_{By} = x/L$$

***( 0<x<a)***

$\eta V_C$ $\qquad \eta V_{CB} = -\eta R_{By} = -x/L$

***(a<x<L)***

$$\eta V_{CA} = \eta R_{Ay} = (L - x)/L$$

$\eta M_C$

***( 0<x<a)*** $\quad \eta M_{CB} = b \cdot \eta R_{By} = b(x/L)$

***(a<x<L)*** $\quad \eta M_{CA} = a \cdot \eta R_{Ay} = a[(L - x)/L]$

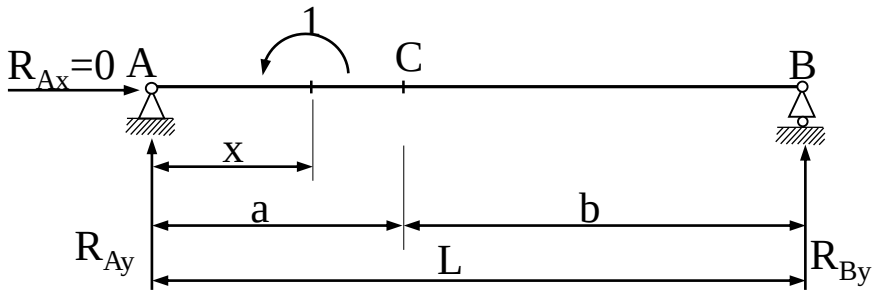
# *Unit moment*

Let's draw the influence lines for

- support reactions
- shear force at point C
- bending moment at point C

of a simply supported beam.

# Influence line of vertical reactions of support A and B

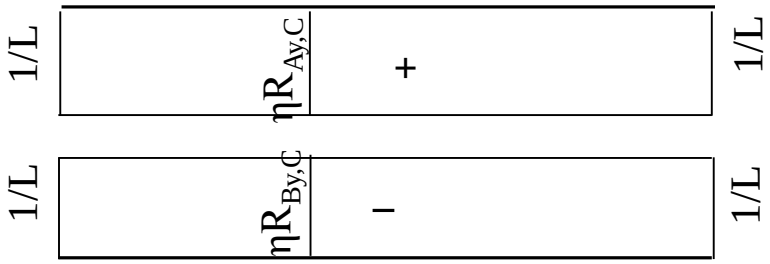



$$\sum M_B = 0 \qquad R_{Ay} = 1/\mathrm{L}$$

Influence line function of vertical reaction of support A: $\boldsymbol{\eta R_{Ay} = 1/L}$

$$\sum M_A = 0 \qquad R_{Ay} = -1/\mathrm{L}$$

Influence line function of vertical reaction of support B: $\boldsymbol{\eta R_{By} = -1/L}$



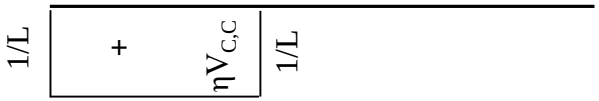

$\eta R_{Ay}$ $\quad \eta R_{Ay} = 1/\mathrm{L}$

$\eta R_{By}$ $\quad \eta R_{By} = -1/\mathrm{L}$

# Influence line of shear force at C

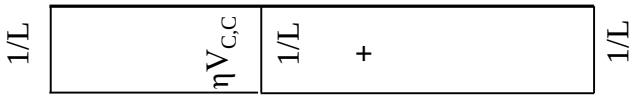
*The load is between A and C (0<x<a)*

$$\eta V_{CB} = -\eta R_{By} = 1/L$$

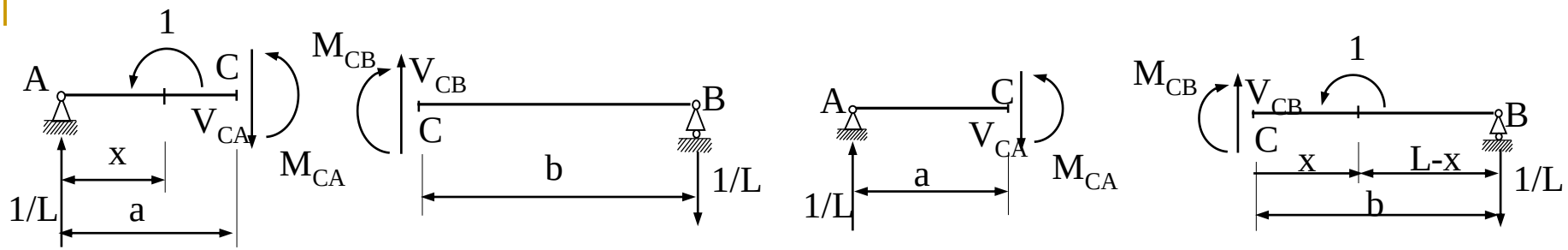



*The load is between C and B (a<x<L)*

$$\eta V_{CA} = \eta R_{Ay} = 1/L$$



# Influence line of bending moment at C




*The load is between A and C (0<x<a)*

Consider CB part since there is less load acting on CB

$$\sum M_C = 0 \qquad M_{CB} = b.R_{By} = \mathrm{b}.(-\frac{1}{L})$$

Influence line function of bending moment at point C:

$$\eta \boldsymbol{M_{CB} = b.\eta R_{By} = -b/L}$$

*The load is between C and B (a<x<L)*

Consider AC part since there is less load acting on AC

$$\sum M_C = 0 \qquad M_{CA} = a.R_{Ay} = \mathrm{a}.(\frac{1}{L})$$

Influence line function of bending moment at point C:

$$\eta \boldsymbol{M_{CA} = a.\eta R_{Ay} = a/L}$$

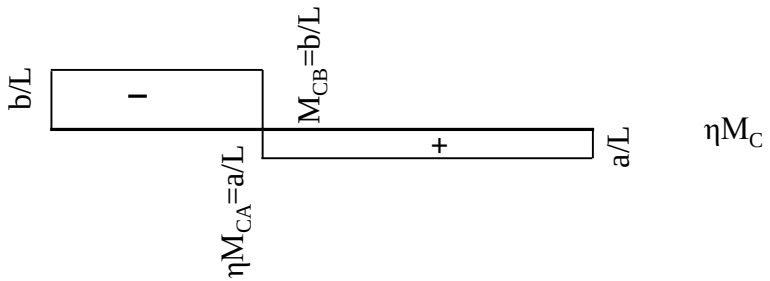
# Influence line of bending moment at C

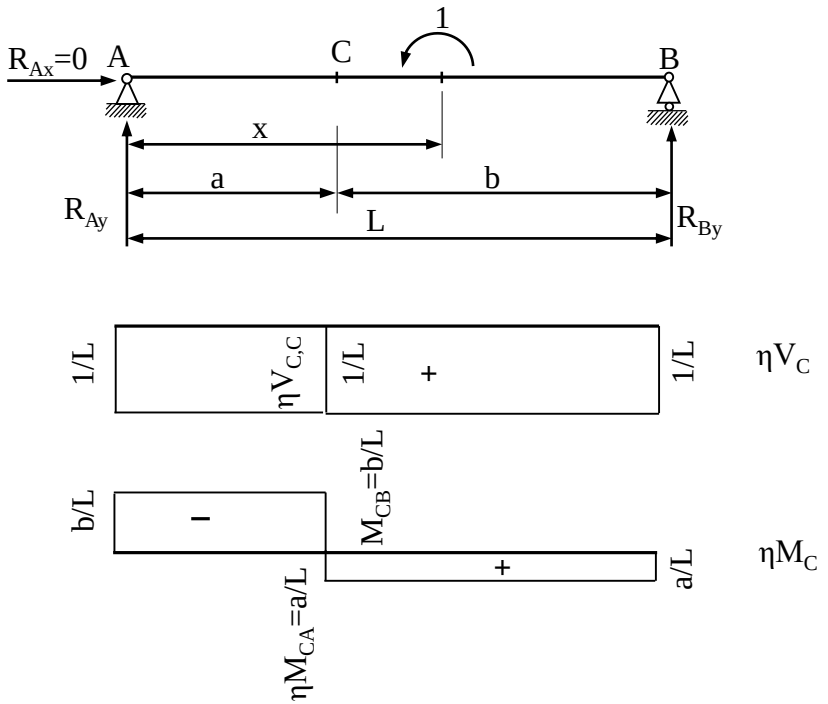
*The load is between A and C (0<x<a)*

$$\eta M_{CB} = b.\eta R_{By} = -b/L$$

*The load is between C and B (a<x<L)*

$$\eta M_{CA} = a.\eta R_{Ay} = a/L$$




1
$R_{Ax}$=0
A
C
B
x
a
b
$R_{Ay}$
L
$R_{By}$
1/L
$\eta V_{C,C}$
1/L
+
1/L
$\eta V_C$
$M_{CB}$=b/L
b/L
−
+
a/L
$\eta M_C$
$\eta M_{CA}$=a/L

## Unit vertical load

- Support reactions
- Shear force
- Axial force

The influence lines are dimensionless

- Moment support reaction
- Bending moment

The units of influence line are m

## Unit moment

- Support reactions
- Shear force
- Axial force

The units of influence line are 1/m

- Moment support reaction
- Bending moment

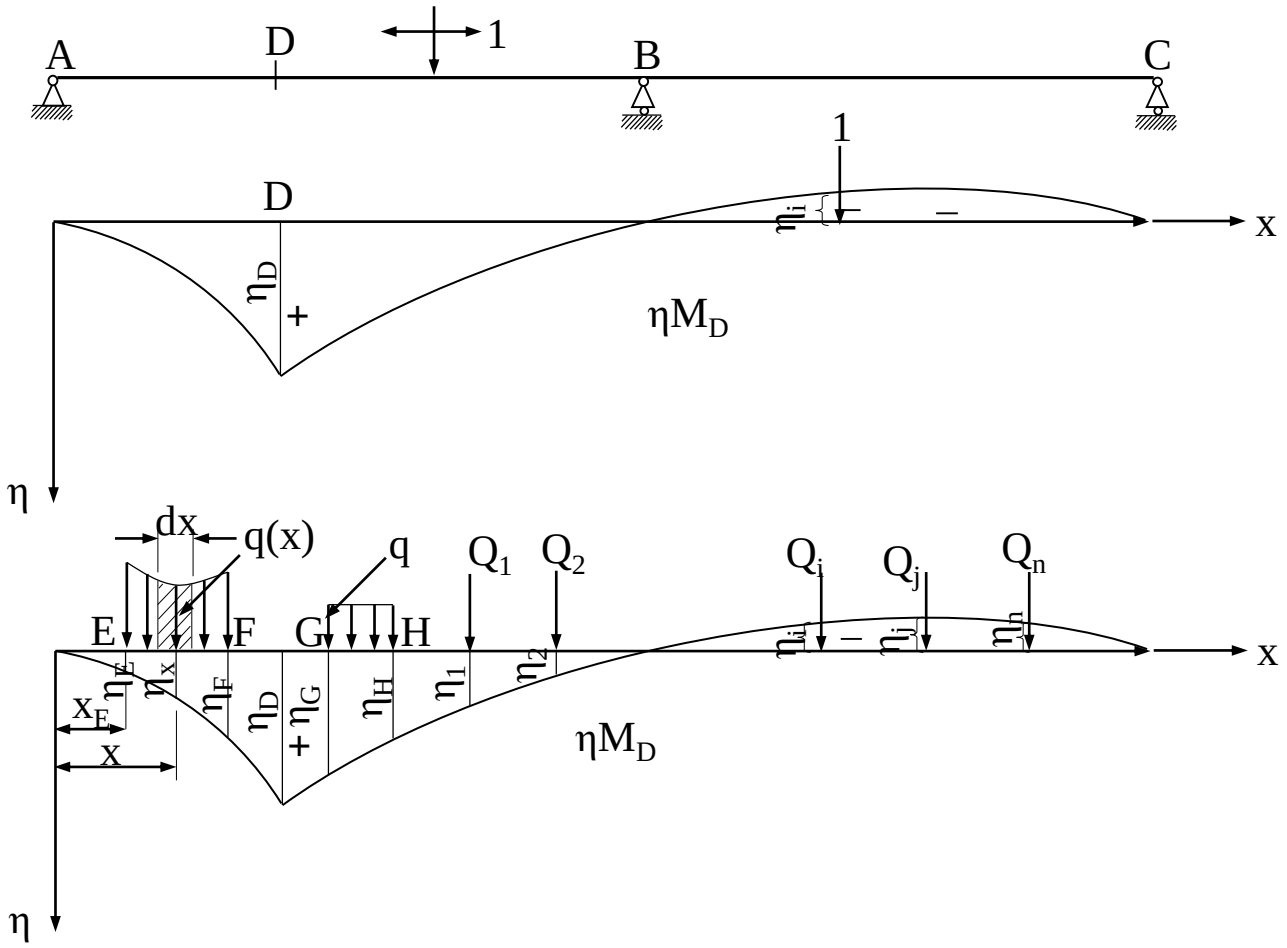
The influence lines are dimensionless

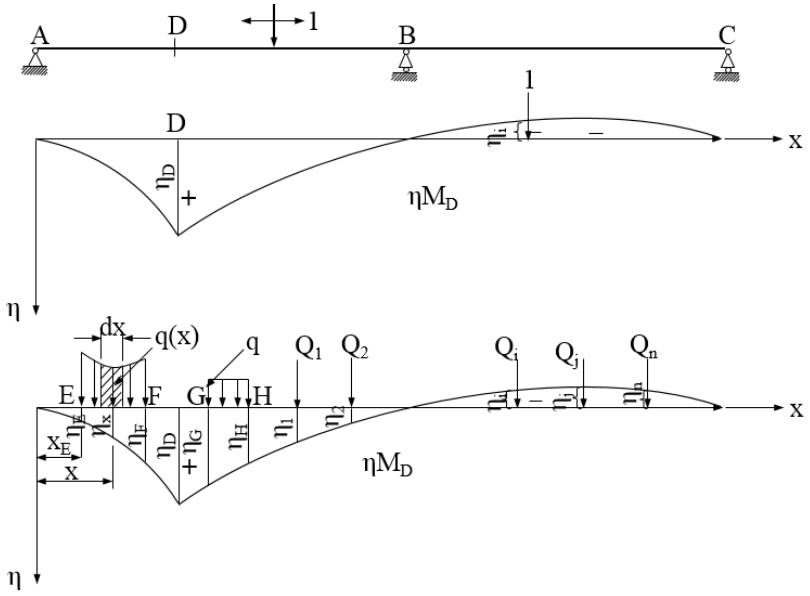
# How to Use Influence Lines

Influence lines can be used to

- Determine support reactions, internal forces, deformations due to dead and live loads
- Determine the location of a live load that makes the support reaction, internal force, deformation at a specific point reach its maximum or minimum value and this maximum and minimum value itself

Consider a two-span continuous beam with the influence line for bending moment at D given in the figure:






- **The values $\eta_i$ are the values of bending moment at D when the unit load is at i.**

Then we can say that the bending moment at D due to $Q_i$ is equal to $\eta_i Q_i$ and the bending moment at D due to point loads at i = 1, 2, ……, n is equal to

$$M_D = \eta_1 Q_1 + \ldots\ldots\ldots + \eta_i Q_i + \ldots\ldots\ldots + \eta_n Q_n$$

$$M_D = \sum_{i=1}^{n} \eta_i Q_i$$

The bending moment at D due the distributed load acting over a length of dx equal to

$$dM_D = q(x).dx.\eta_x$$

Accordingly, the bending moment at d due to the distributed load acting over EF is equal to

$$M_D = \int_{x_E}^{x_F} q(x)\eta_x dx$$

Then the bending moment at D due to the uniform distributed load acting over GH is equal to

$$M_D = \int_{x_G}^{x_H} q(x)\eta_x dx = q\int_{x_G}^{x_H} \eta_x dx = qA_\eta$$

where $A_\eta = \int_{x_G}^{x_H} \eta_x dx$

# Reference:

All the figures are taken from the book given below:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ